

MGMT3308

Supply Chain Management

Week 4: Inventory Management

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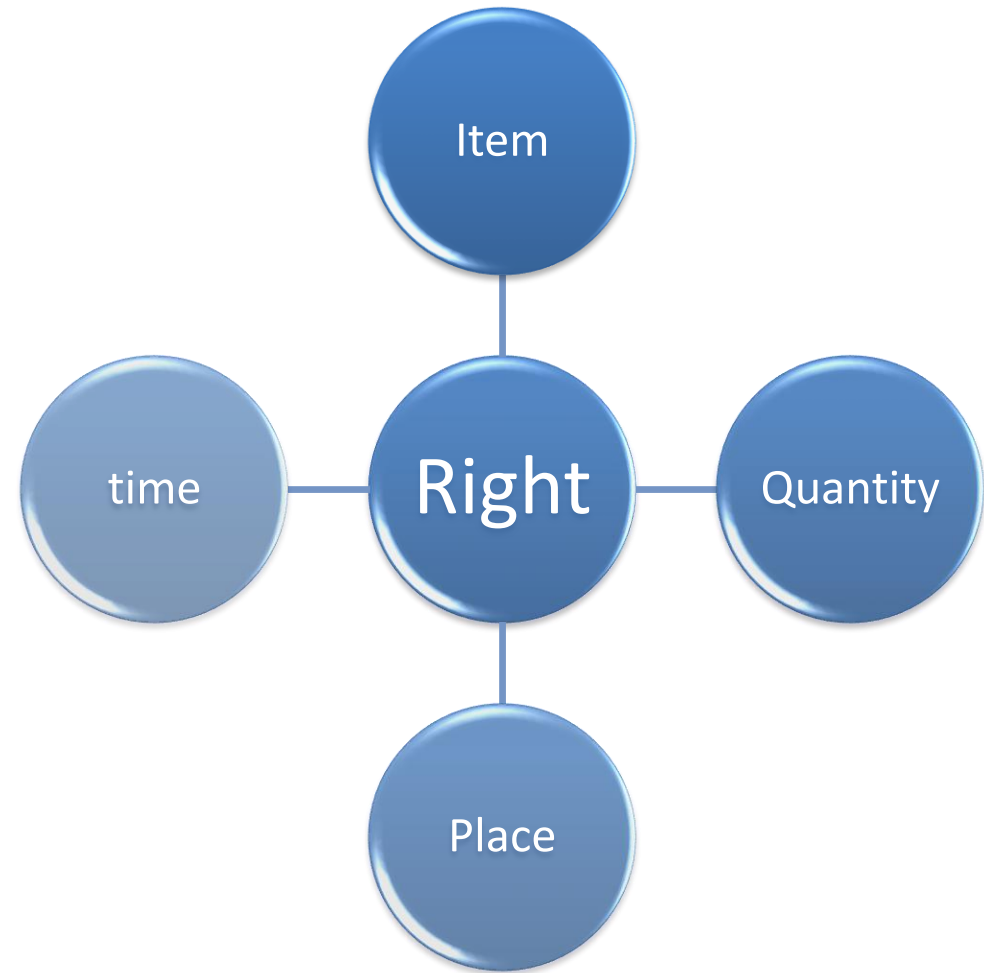
What is inventory management?

- Inventory refers to goods and materials held for sale, production, or service.
- **Inventory management** is the process of **ordering, storing, tracking, and controlling** a company's inventory — the goods and materials it holds for sale, production, or service.





Inventory Management



Inventory management

Right item



Right Item – Make sure the correct product or material is stocked, matching customer or production needs.

Inventory management



Right Place



Right time
(Availability)

Inventory
management

Inventory management

Right time
(Lead time)



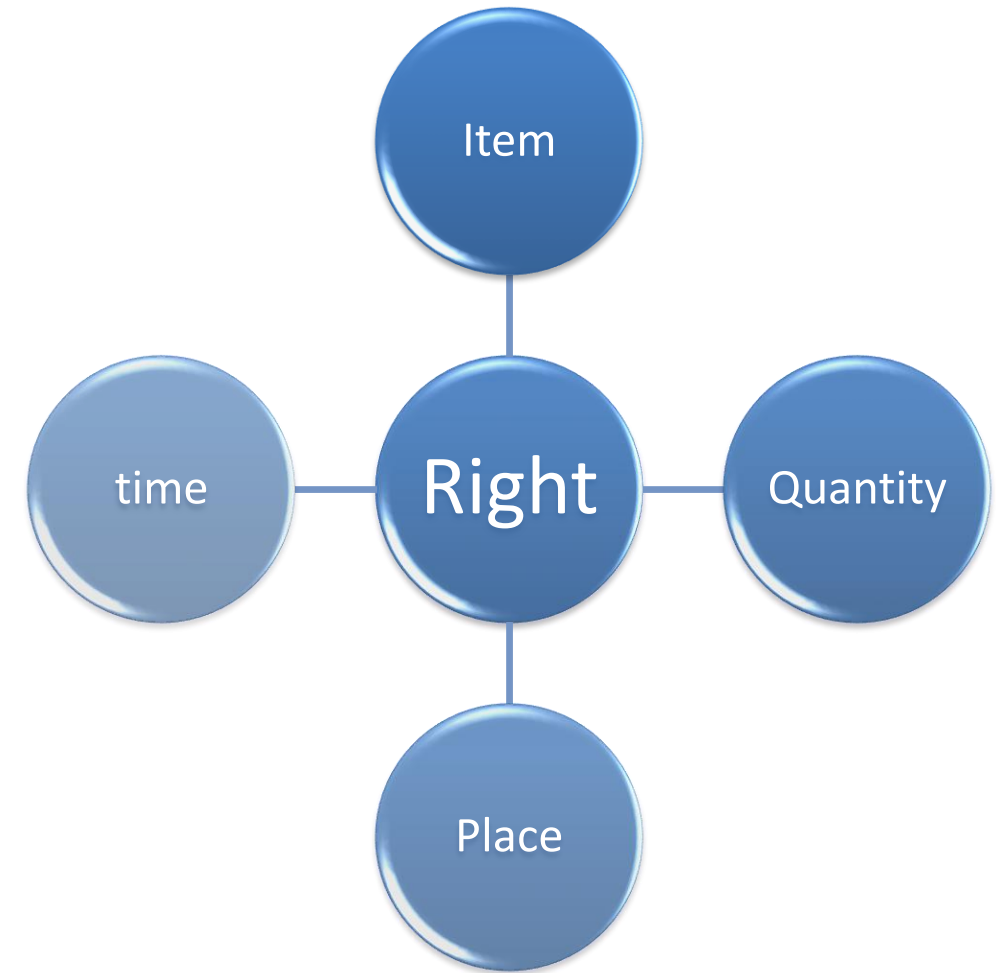
Right quantity

Inventory
management

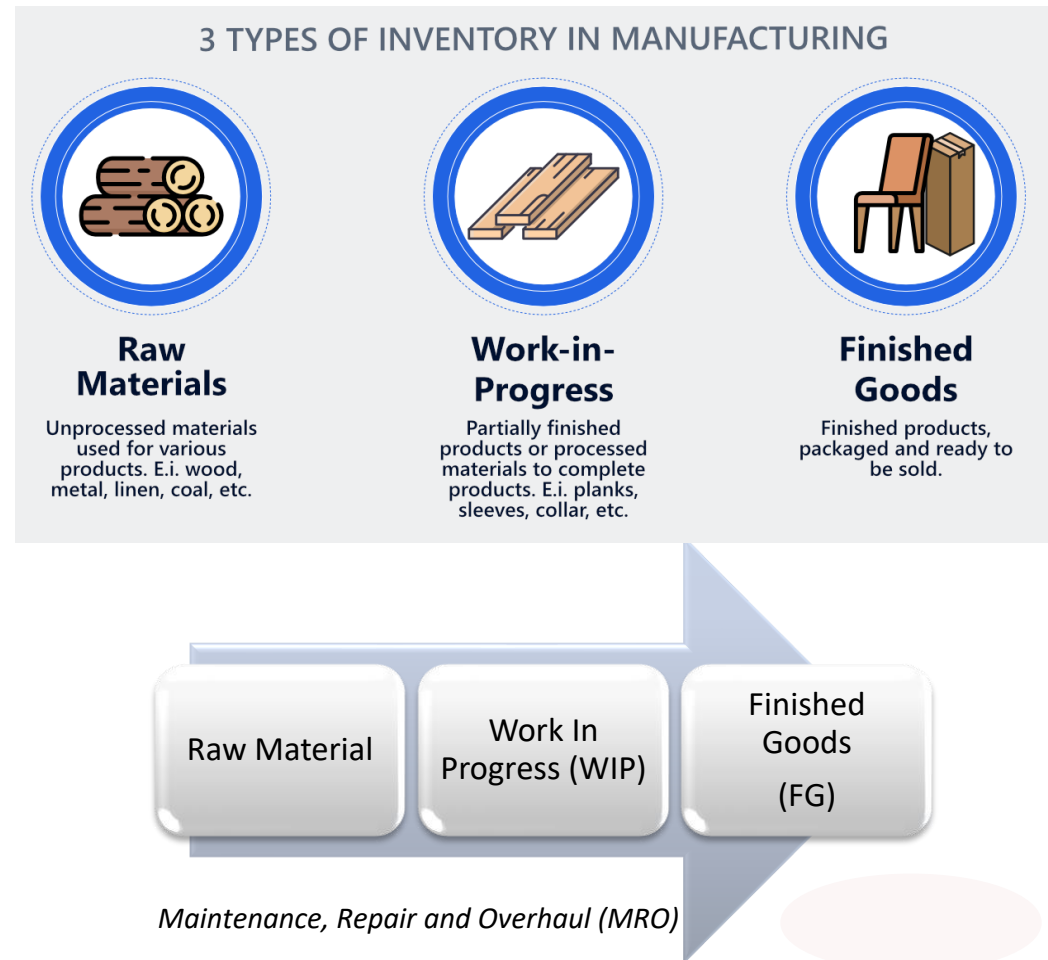




Inventory Management



Types of Inventories



Types of Inventories

Example:

Balance Sheet of
General Motor

Annual Financials for General Motors Co.

[+ View Ratios](#)

Assets

Fiscal year is January-December. All values USD millions.

	2015	2016	2017	2018	2019	5-year trend
+ Cash & Short Term Investments	24.93B	25.8B	25.57B	28.89B	26.6B	
Cash Only	16.77B	13.96B	17.26B	22.93B	22.42B	
Short-Term Investments	8.16B	11.84B	8.31B	5.97B	4.17B	
+ Total Accounts Receivable	26.39B	24.83B	28.69B	33.4B	33.4B	
Accounts Receivables, Net	26.39B	24.83B	28.69B	33.4B	33.4B	
Accounts Receivables, Gross	26.72B	25.04B	28.96B	33.61B	33.6B	
Bad Debt/Doubtful Accounts	(327M)	(212M)	(278M)	(211M)	(201M)	
Other Receivables	-	-	-	-	-	
Inventories	13.76B	11.04B	10.66B	9.82B	10.4B	
Finished Goods	8.62B	6.03B	6.46B	5.54B	5.69B	
Work in Progress	-	-	-	-	-	
Raw Materials	5.15B	5.01B	4.2B	4.27B	4.71B	

<https://www.marketwatch.com/investing/stock/gm/financials/balance-sheet>

Types of Inventories

Example

Balance Sheet of Amazon

Quarterly Financials for Amazon.com Inc.

[+ View Ratios](#)

Assets

All values USD millions.

	31-Mar-2019	30-Jun-2019	30-Sep-2019	31-Dec-2019	31-Mar-2020	5-qtr trend
+ Cash & Short Term Investments	37.38B	41.78B	43.7B	55.34B	49.57B	
Cash Only	23.47B	22.93B	23.56B	36.41B	27.48B	
Short-Term Investments	-	-	-	-	-	
+ Total Accounts Receivable	15.62B	16.43B	16.59B	20.5B	17.55B	
Accounts Receivables, Net	15.62B	16.43B	16.59B	20.5B	17.55B	
Accounts Receivables, Gross	15.62B	16.43B	16.59B	21.21B	18.65B	
Bad Debt/Doubtful Accounts	-	-	-	(718M)	(1.1B)	
Other Receivables	-	-	-	-	-	
Inventories	16.43B	18.58B	18.77B	20.5B	18.86B	
Finished Goods	16.43B	18.58B	18.77B	20.5B	18.86B	
Work in Progress	-	-	-	-	-	
Raw Materials	-	-	-	-	-	

<https://www.marketwatch.com/investing/stock/amzn/financials/balance-sheet/quarter>



Manager's
Special
LEMON

\$2.99

LB

Manager's
Special
ORANGE-SOUR

\$2.99

LB

Manager's
Special
YAM-COSTA RICA
\$1.99

LB

New Lower Price
YAM-JUMBO
79¢

LB

Manager's
Special
CABBAGE-CHINESE
\$1.29

LB

Manager's
Special
CABBAGE-SAVOY
\$1.29

LB

Manager's
Special
CABBAGE-RED
\$1.29

LB

New Lower Price
CABBAGE-GREEN
79¢

LB

Manager's
Special
CABBAGE-GREEN
79¢

LB



INVENTORY MANAGEMENT

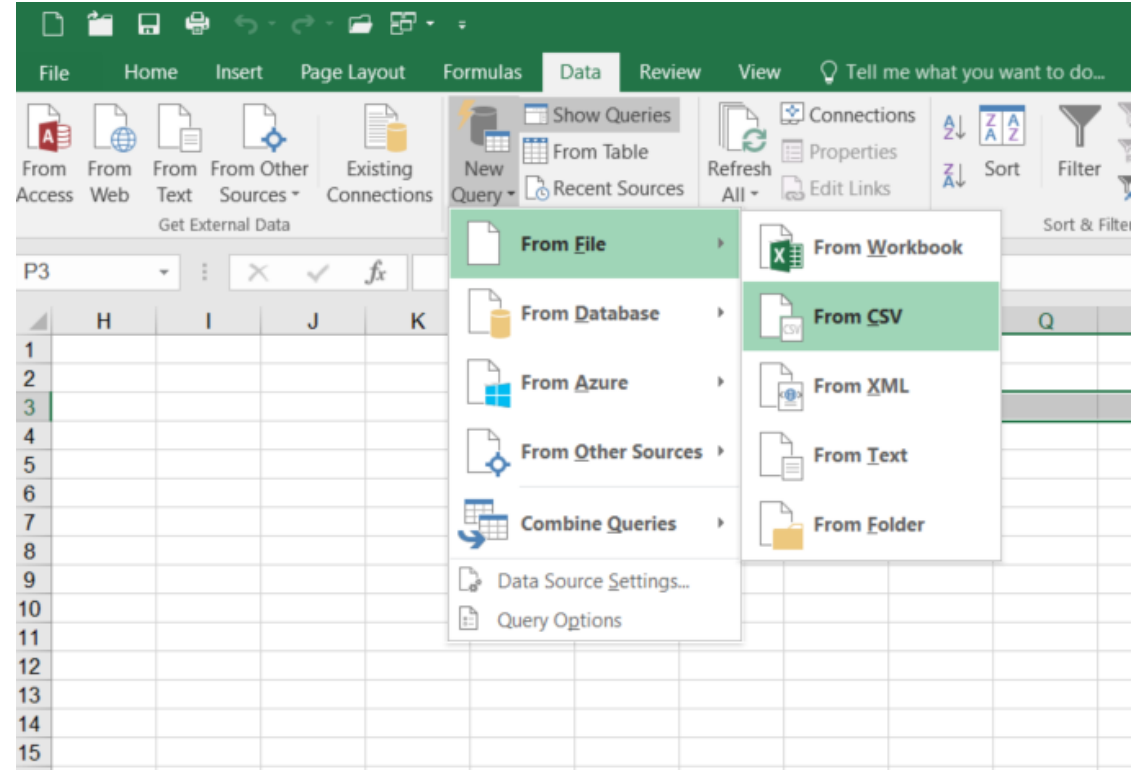
Inventory is another name for materials and is any material that a firm holds in order to **satisfy customer demand**.

When the raw materials are processed, but are not yet completely finished, they are called **work in progress**.

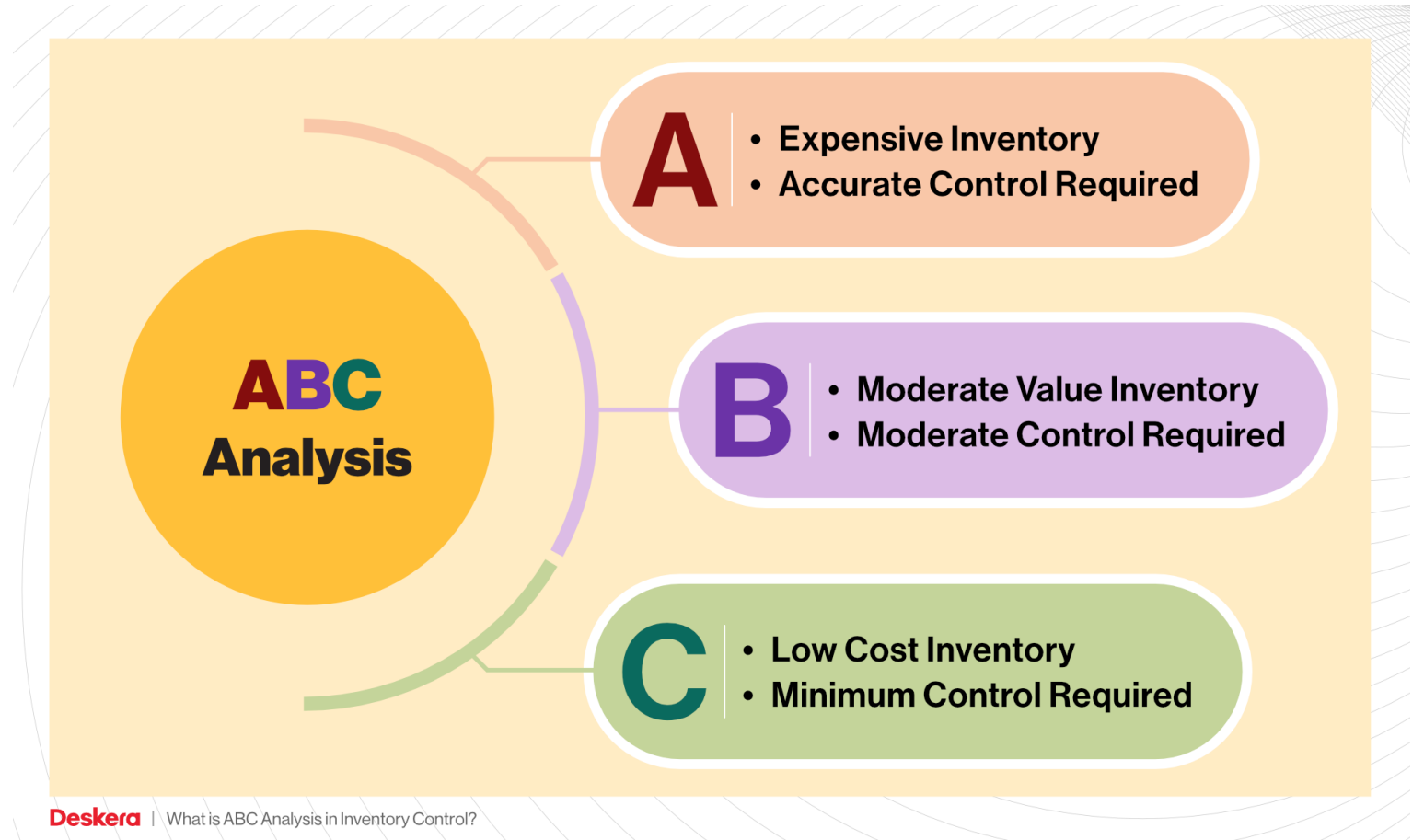
- Inventory hides problems
- Inventory costs!
- The money could have been invested elsewhere.



Importing data from difference sources



ABC analysis



ABC analysis

It can be seen that the top two items (# 373 and # 539) account for just over 65% of the expense.

'A' items may be controlled closely, using the reorder point system; the less demanding periodic system may be used for 'B' items; and 'C' items may be blanket purchased once or twice in a year.

Item	Annual Expenses	Percentage of Total	Classifications
373	46,335	45.77%	A
539	19,611	19.37%	A
455	8007	7.91%	B
769	6181	6.11%	B
441	5526	5.46%	B
65	5503	5.44%	B
205	3278	3.24%	C
401	3063	3.03%	C
352	2845	2.81%	C
543	603	0.60%	C
454	179	0.18%	C
432	111	0.11%	C
Total	101,242	100.00%	



ABC analysis

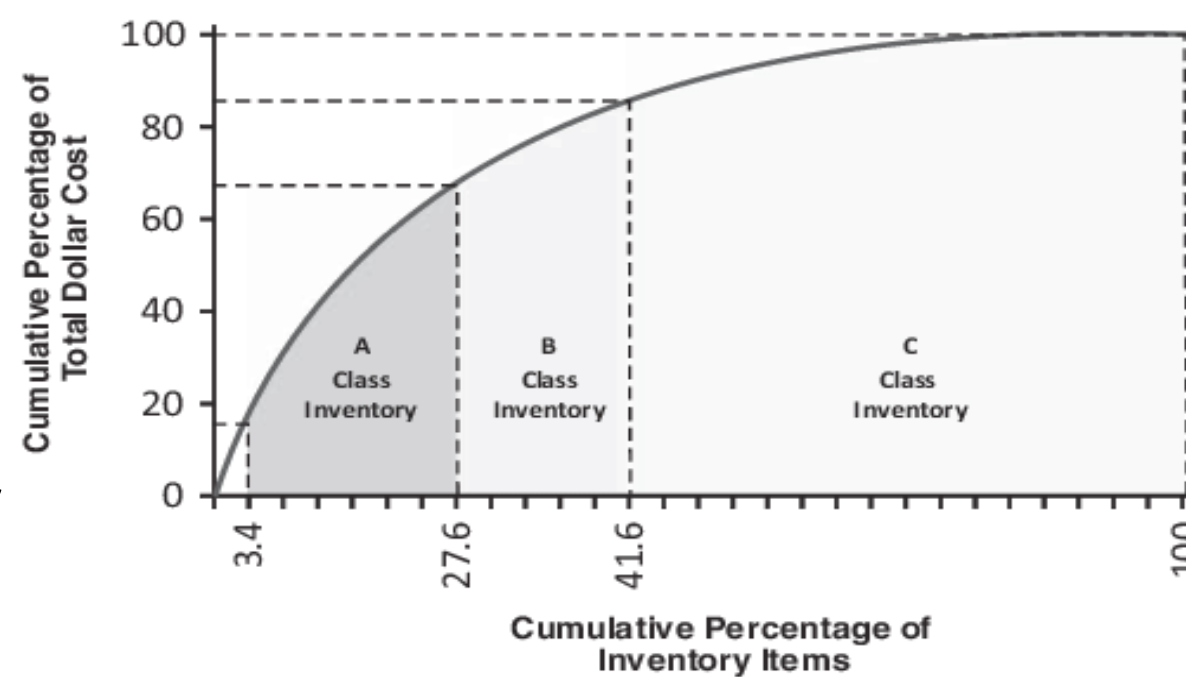
This is derived from the 'Pareto' or '80/20' rule

Most firms have far too many inventory items to manage. They often use a tool called [ABC analysis](#) to separate out the most important items so that more attention can be focused on those items.

A: Engine major parts, transmission systems, fuel injection system

B: Windshield wipers, air conditioning compressors, exterior mirrors, floor mats, mudguards, roof racks

C: Screws, basic tools, gloves, cleaning solvents, toilet tissues, printer papers, tapes



INVENTORY REDUCTION PRINCIPLES

- **Pool inventory**

This is the case in inventory centralisation where *demand from different locations is combined*, or in *delayed product differentiation* where demand for different products is combined, or by using *common components* where demands for different components are combined.

- **Reduce variation**

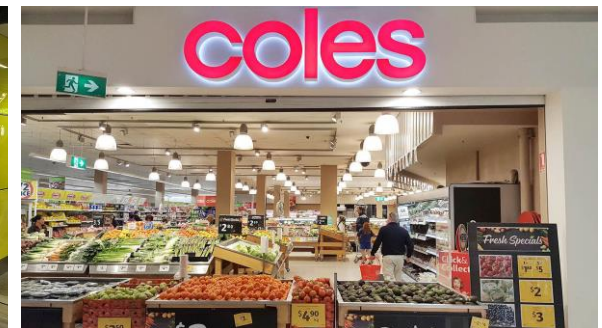
Recall that the reason for holding safety stock is variation. *Variation of lead time, variation of demand, variation of supply, variation of quality*, all contribute to safety stock.

- **Reduce lead time**

Lead time directly affects inventory held. For example, the reorder point formula shows that the ROP can be reduced if the lead time can be reduced. Likewise, transit inventory costs can be reduced by reducing the lead time.

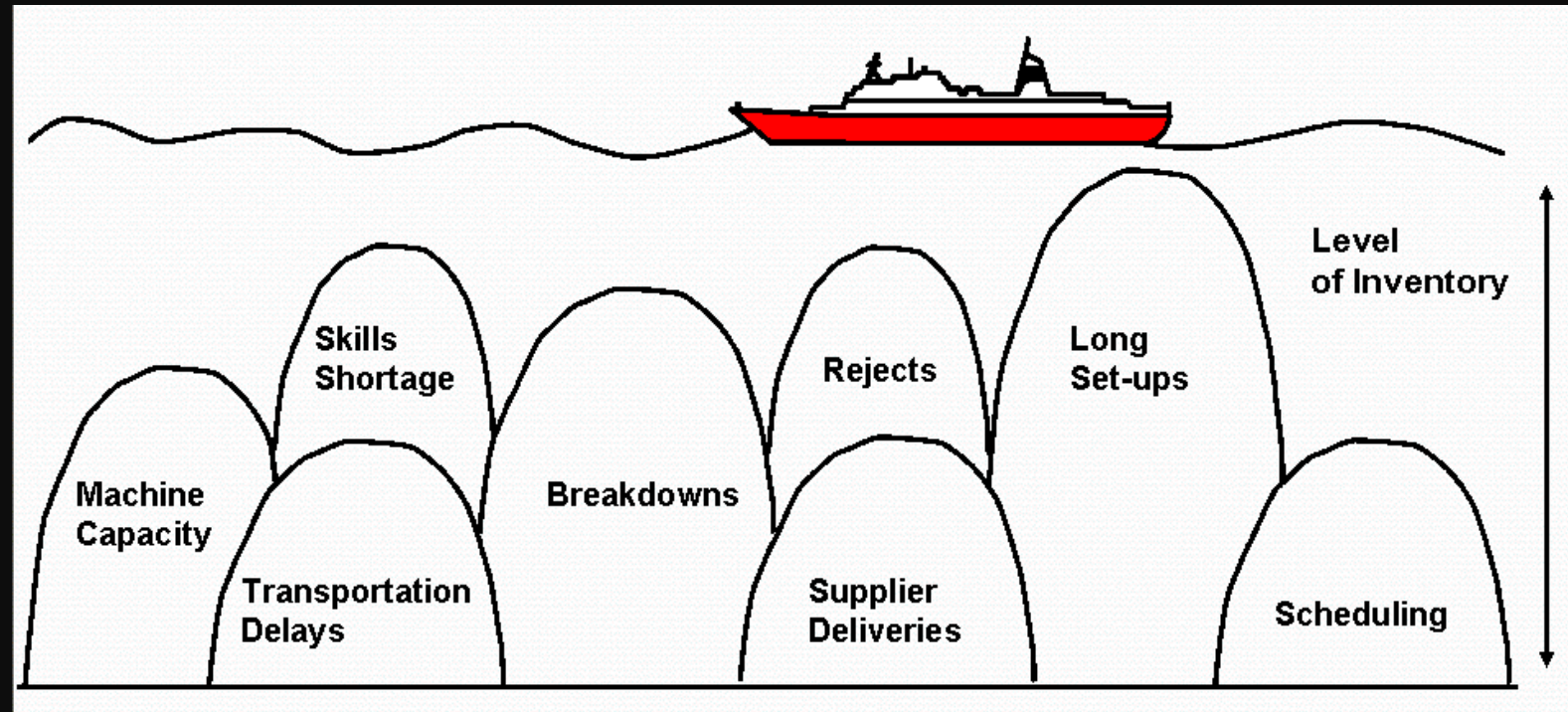
- **Just-in-time inventory (JIT) philosophy**

JIT is as much a *philosophy* as it is a technique.



Inventory covers issues

- Real World Examples: Toyota
- Apple

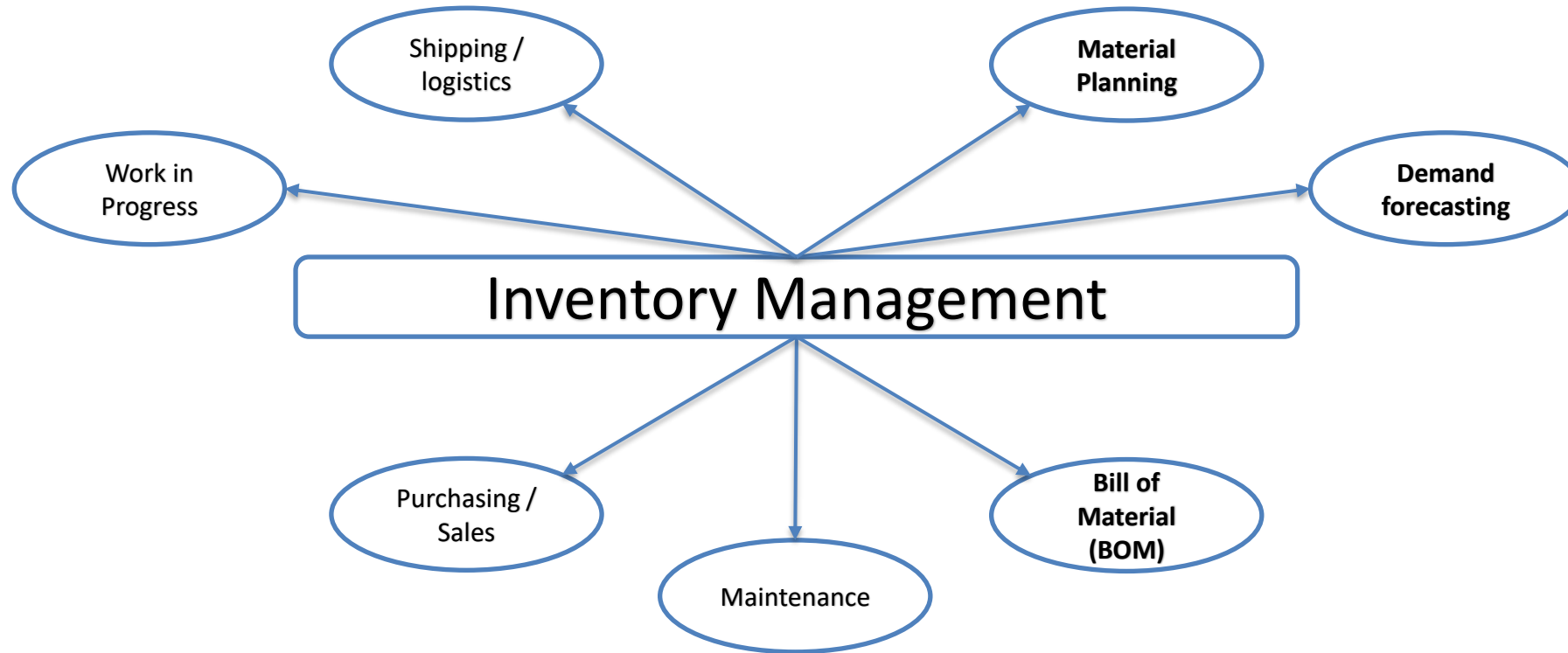




Opportunity cost (lost on carrying inventories)

- *This is the amount of money the firm would have earned if the money were invested elsewhere other than in inventory.*

The BIG Picture



Sustainability Concerns

Change of customer tastes and preferences

Power and resources used to carry inventories

Example (from the fashion industry):

In 2018, Burberry burned down almost \$30m worth unsold goods



BBC

<https://www.bbc.com/news/business-44885983>

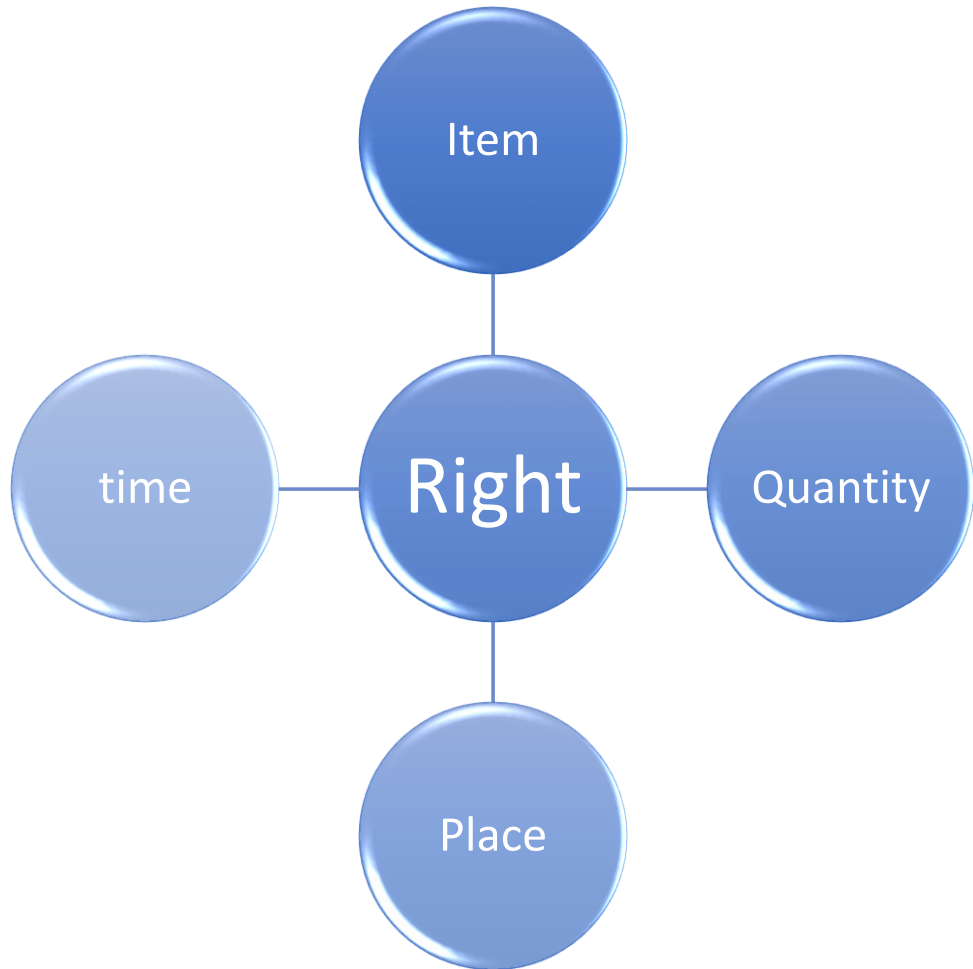
Burberry burns bags, clothes and perfume worth millions

19 July 2018 — Burberry, the upmarket British fashion label, **destroyed unsold clothes, accessories and perfume worth £28.6m last year** to protect its brand. It ...

Key Concepts

BIG

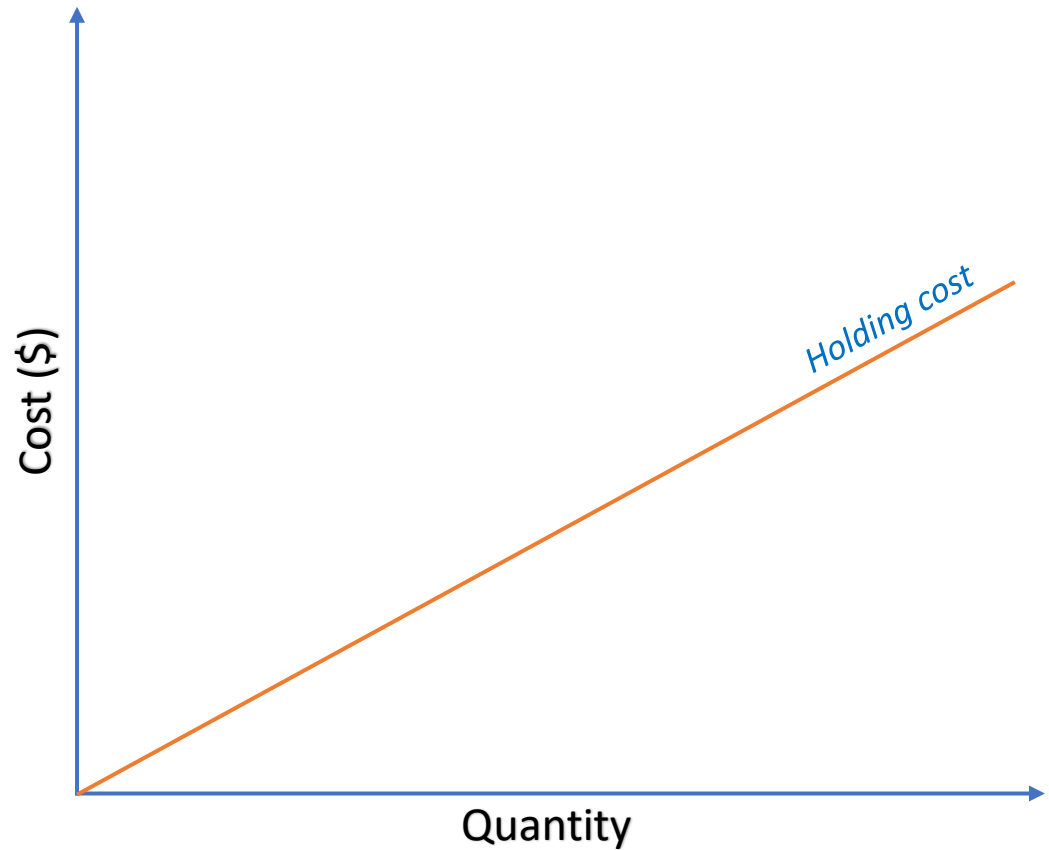
inventories?



Key Concepts

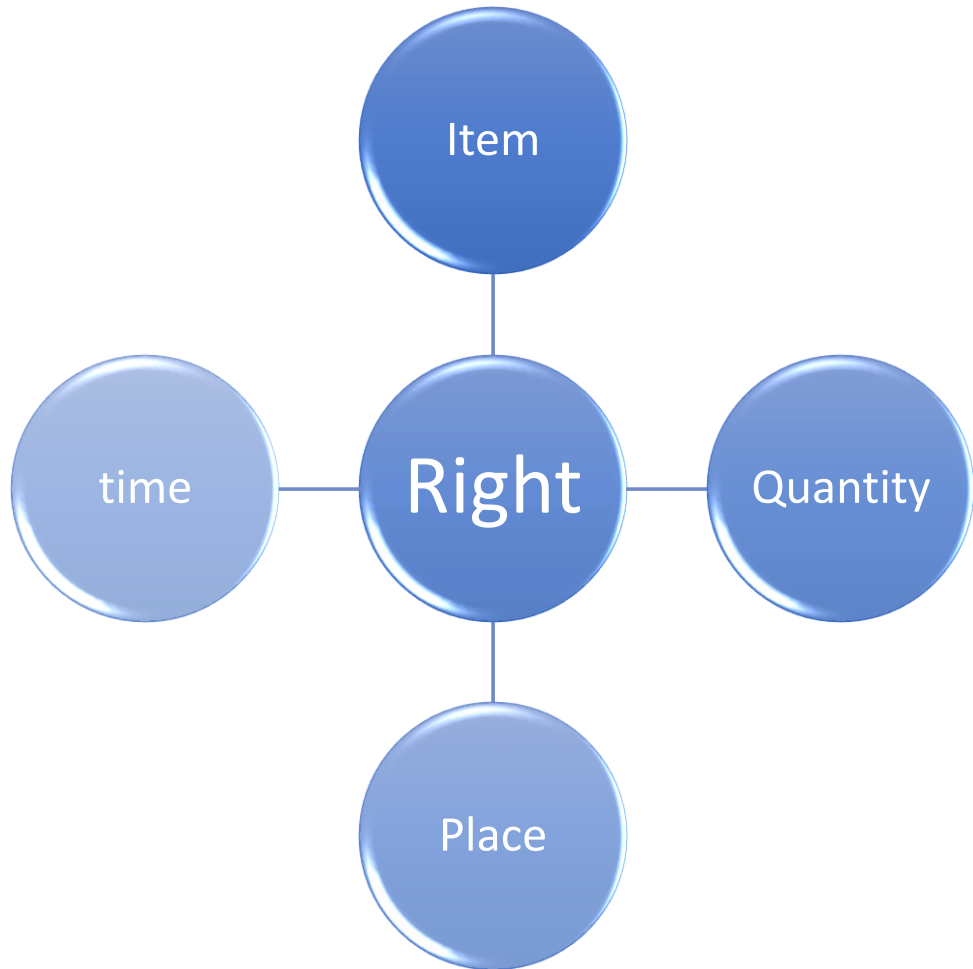
Holding cost

- Storage
- Insurance
- Tax
- Maintenance
- Obsolescence (expiring)
- Opportunity costs



Key Concepts

*Small
inventories?*



Key Concepts

Order cost

- **Administrative Costs:**

Paperwork, communication, and processing the order.

- **Transportation Costs:**

Shipping or delivery fees charged by suppliers.

- **Setup Costs:**

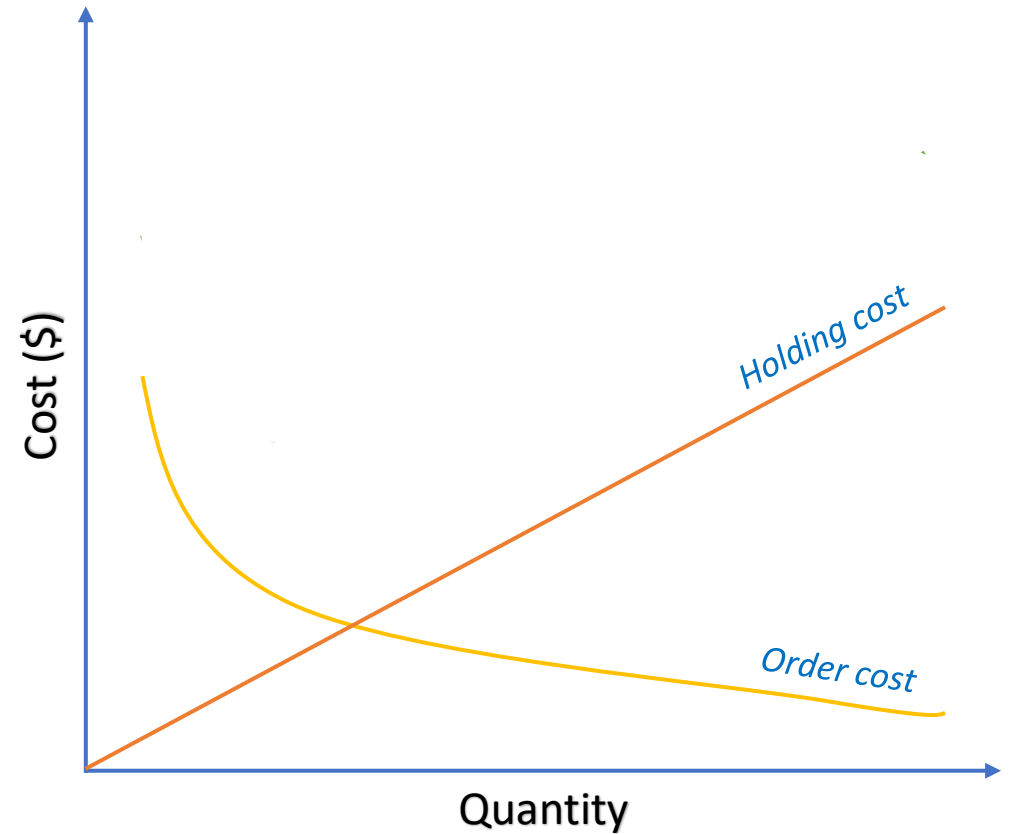
Time and resources needed to prepare for a new order, such as production line adjustments if applicable.

- **Miscellaneous Costs:**

Additional fees like handling charges or procurement system costs.

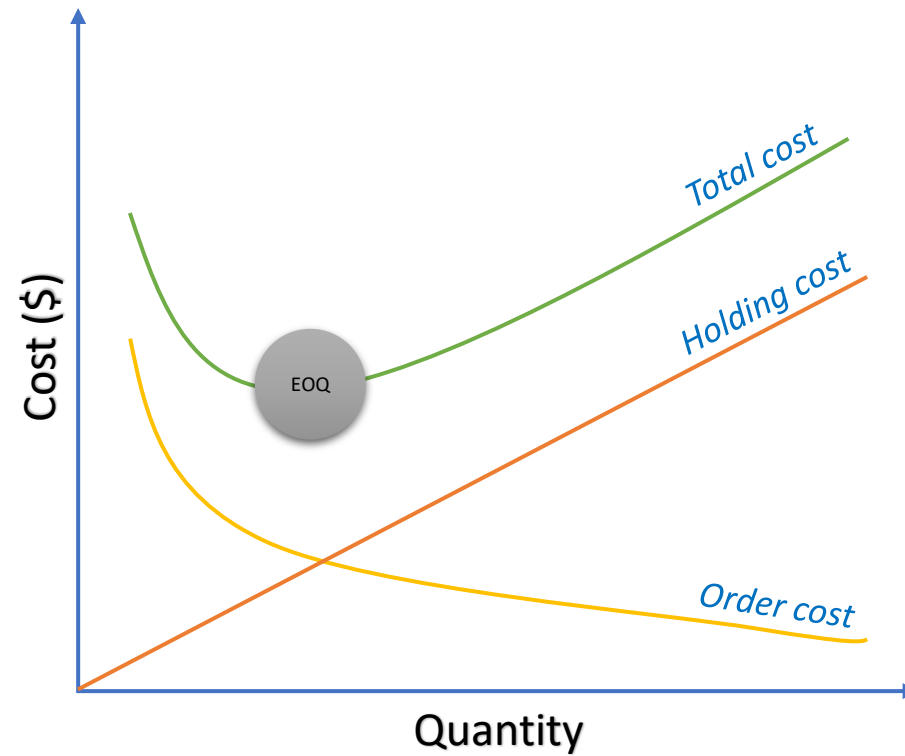
- **Lost sale:**

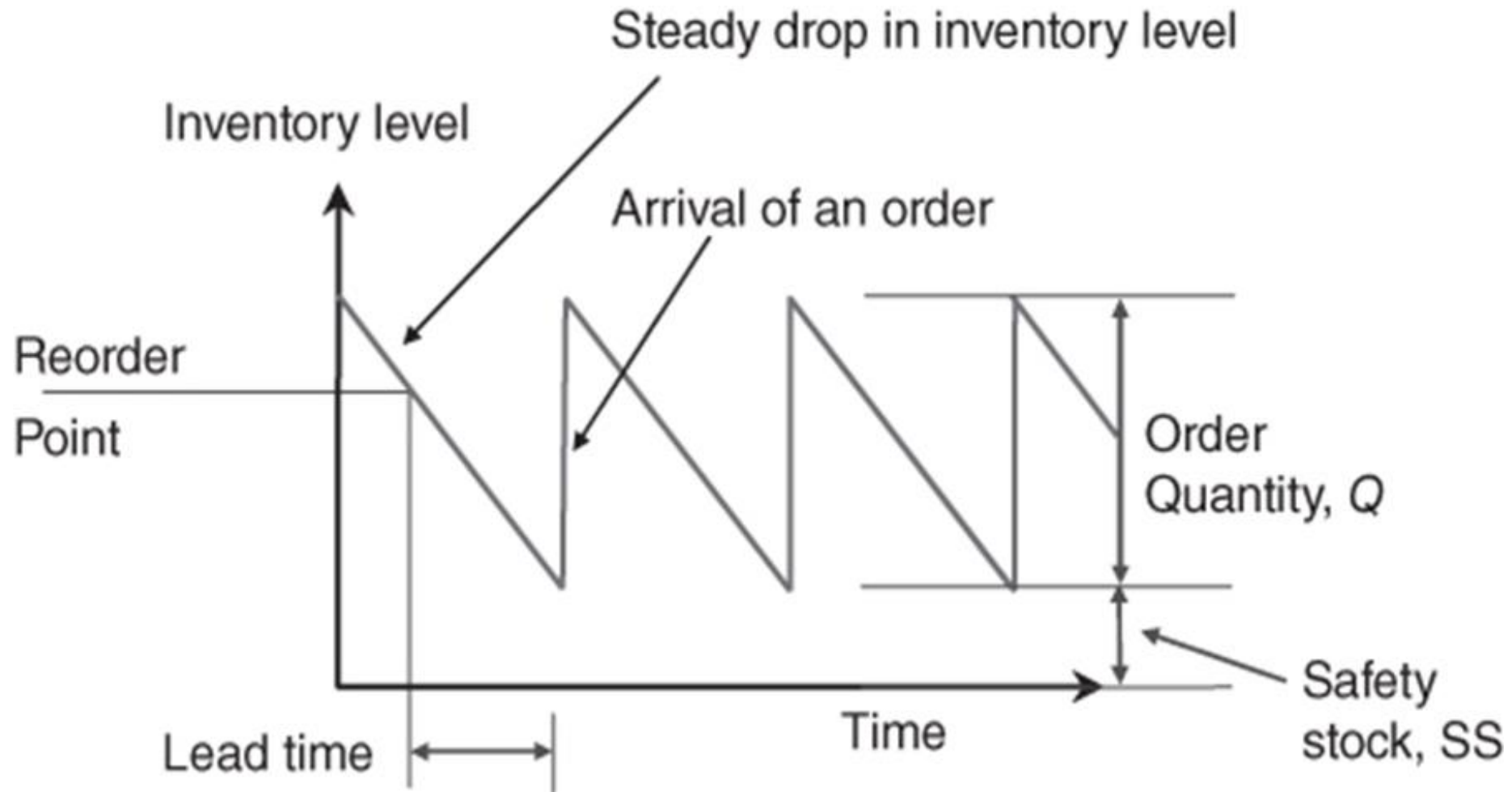
This can also sometimes result in permanent loss of some customers!



Key Concepts

- Economic Order Quantity EOQ

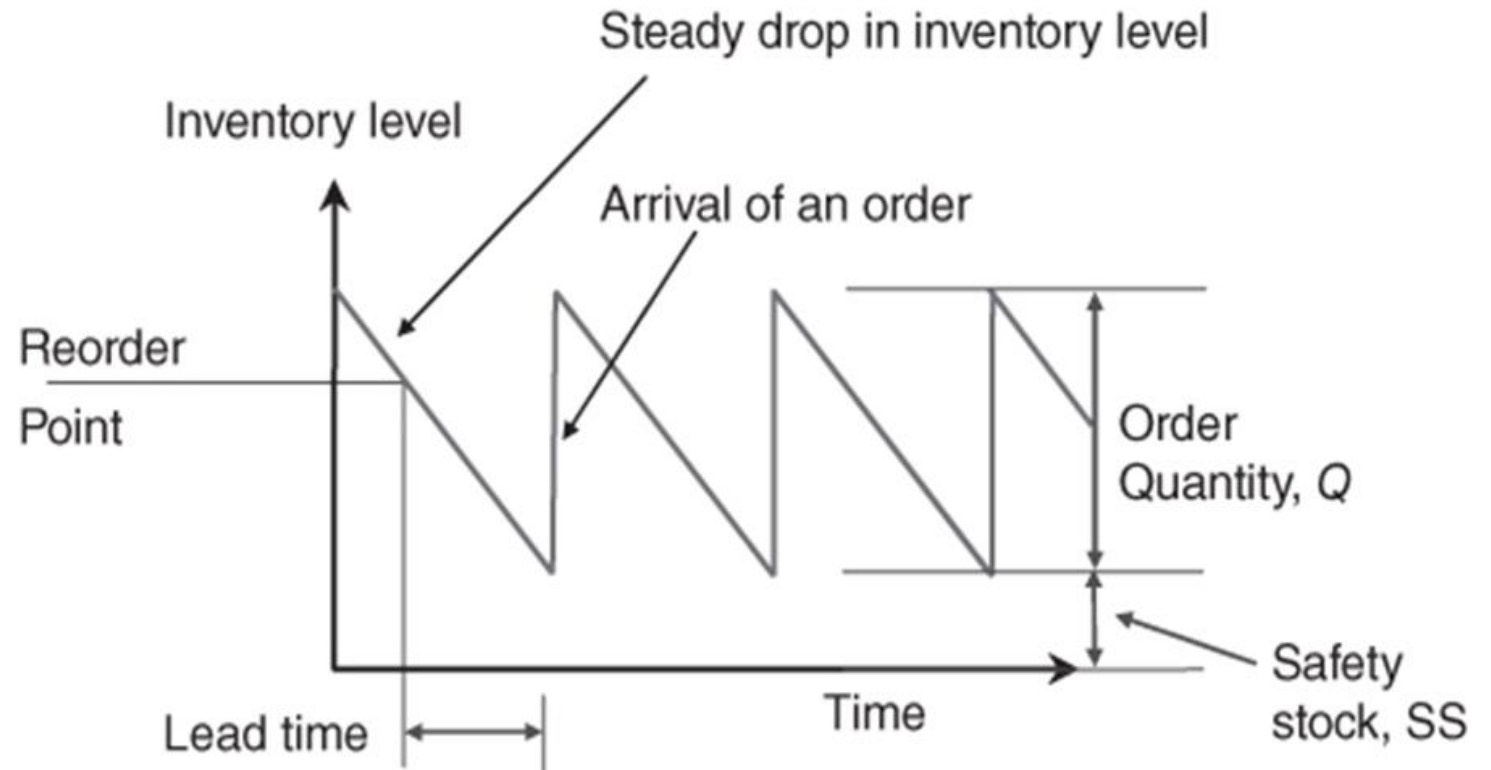




EOQ Model: Basic Definitions

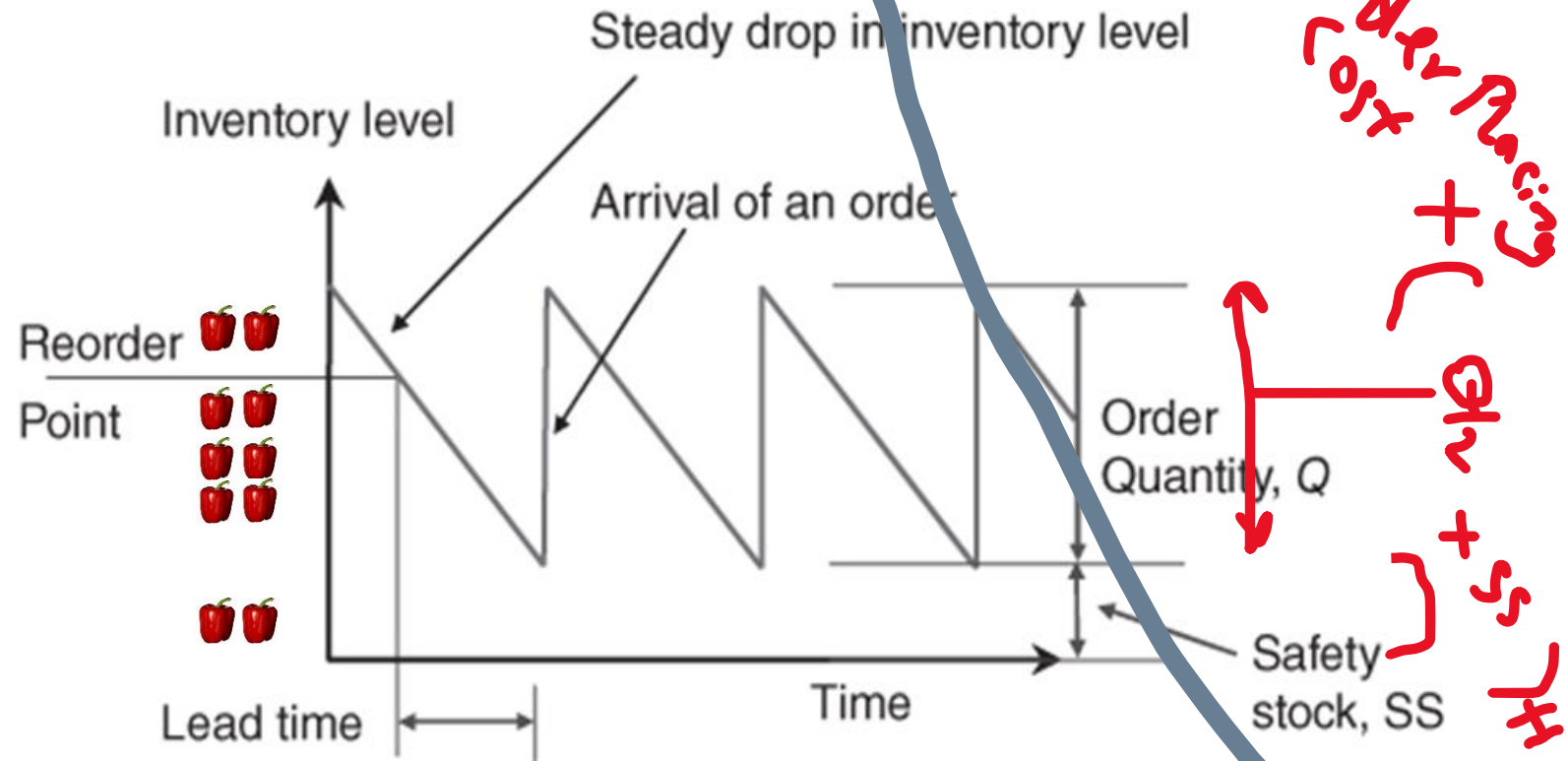
Total cost of inventory

- **Average inventory level** $= (SS + Q + SS) / 2 = SS + Q/2$
- And **the annual holding cost** $= (SS + Q/2)H$
- If S is the order processing cost per order:
- **Annual order processing cost** $= (D/Q)S$
- Adding these three costs the total annual inventory costs associated with this item are calculated below.
- **Total annual cost (TAC)** =
- **Purchase cost + Holding cost + Order processing cost** =
- $(SS + Q/2) \text{Price} + (SS + Q/2) \text{Holding Cost} + (D/Q) \text{Set Up Cost}$



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EOQ

It is straightforward to derive the EOQ formula. Differentiating the expression for TAC and setting to zero for minimisation:

$$\begin{aligned}\frac{d(TAC)}{dQ} &= 0, \\ \Rightarrow \frac{d\left(p \times D + \left(SS + \frac{Q}{2}\right)H + \frac{D}{Q}S\right)}{dQ} &= 0 \\ \Rightarrow \frac{H}{2} - \frac{D \times S}{Q^2} &= 0\end{aligned}$$

Assuming that the purchasing cost is constant (no bulk discounts). We also assume that the safety stock remains fixed as order quantity is changed:

$$\Rightarrow Q = \sqrt{\frac{2DS}{H}}$$

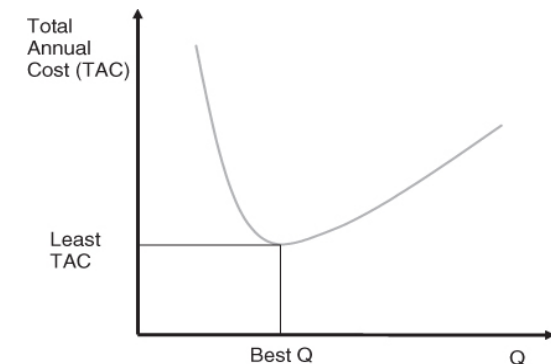
EOQ: Economic Order Quantity

D = annual demand

H = holding cost

S = ordering cost

$$EOQ = \sqrt{\frac{2 \times D \times S}{H}}$$



EOQ

The Fine Garments Company sells fashion clothing. The forecasted annual demand for its premium leather jacket is 1200. The order processing cost per order is \$25, and inventory holding cost is \$50/item/year. How many leather jackets should it order in one shipment?

Answer:

$$\begin{aligned}EOQ &= \sqrt{\frac{2DS}{H}} \\&= \sqrt{\frac{2 \times 1200 \times 25}{50}} \\&= 34.64 \\&\approx 35\end{aligned}$$

To minimise its annual inventory costs, Fine Garments should request 35 leather jackets each time it places an order with its supplier.

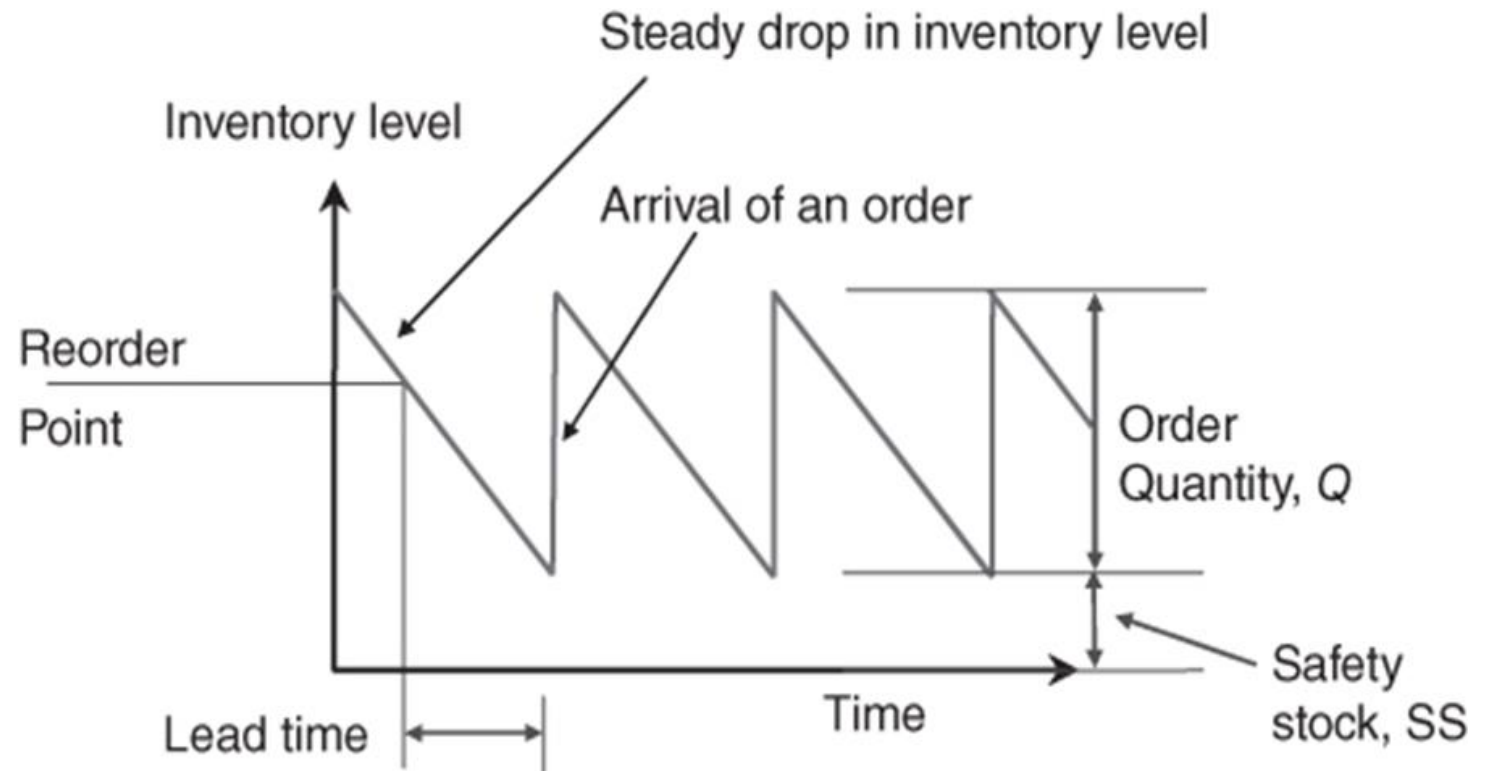
EOQ: Economic Order Quantity

D = annual demand

H = holding cost

S = ordering cost

Safety Stock Formula (Simple Version)



Safety stock
= (maximum use during until the order arrives)
– (average use until the order arrives)



Safety Stock (Buffer)

*This is the amount of inventory **stocked by the system in case of unforeseen events arising**. For example, late **deliveries**, or a sudden increase in **demand**.*

*Other reasons for maintaining safety stock include:
A safeguard against issues such as **poor quality**,
production problems and transportation problems.*

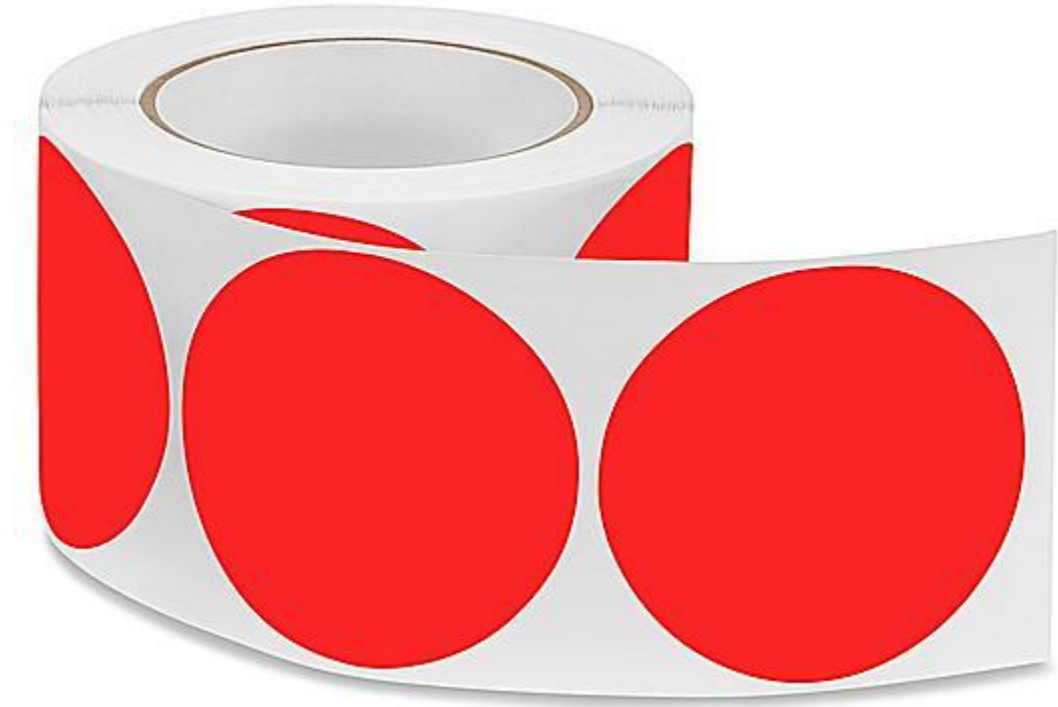
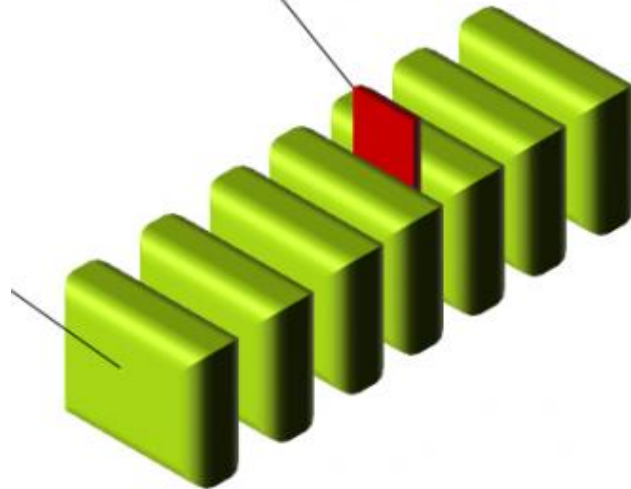
*The root reason for safety stock could be described
as variation – variation of **demand**, variation of **lead time**,
variation of **production rate**, etc.*

*If there was no variation, firms would not need
safety stock.*

Safety stock costs!



REORDER POINT →



Safety Stock

This is the amount of **inventory stocked to cope with uncertainty** (in case of unforeseen events arising).

Examples:

Late deliveries from your suppliers or transportation problems

Higher consumption of inventory than the forecast

Poor quality of some inventories (not usable)

Production problems which may waste some inventories

Transportation problems

In such cases, without safety stock, customer service will suffer.

The **root reason for safety stock could be described as variation** such as:

Variation of **demand**

Variation of **lead time**

Variation of **production**

If there was no variation, firms would not need safety stock.

Safety stock is thus sometimes referred to as buffer stock.

Safety stock is not free!

Inventory centralisation

Example: Automotive Industries

Delayed product differentiation

➤ Principle of Postponement

Example: customising language and countries of smart TVs can be delayed (or be delegated to customers)

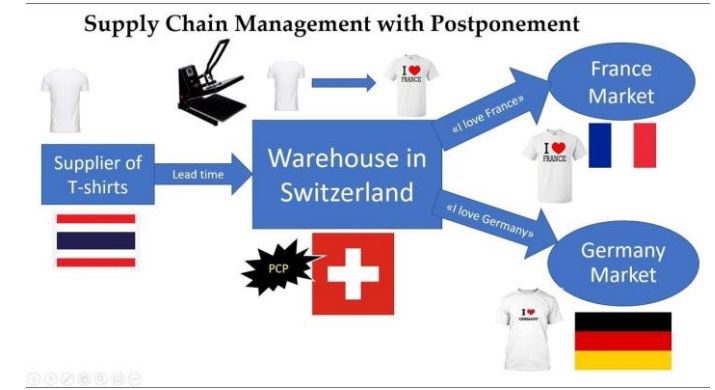
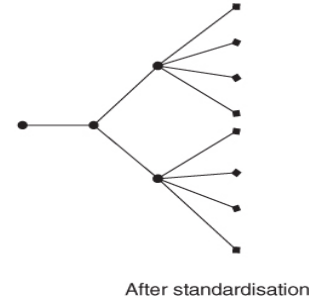
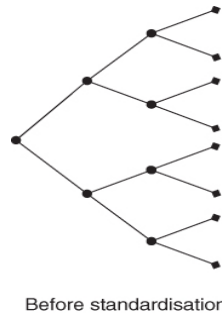
If you are producing sofas, it is good if you can have similar bases to be assembled, so that you can leave the customisation (color, upholstery, and similar) to later stages closer to being shipped to the market.

Increasing Part commonality

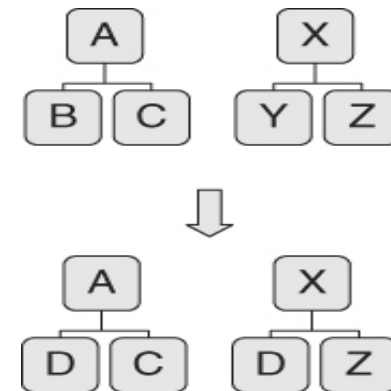
Example: All mobile phones, tablets and cameras sold in the EU have to be equipped with a USB Type-C charging port.

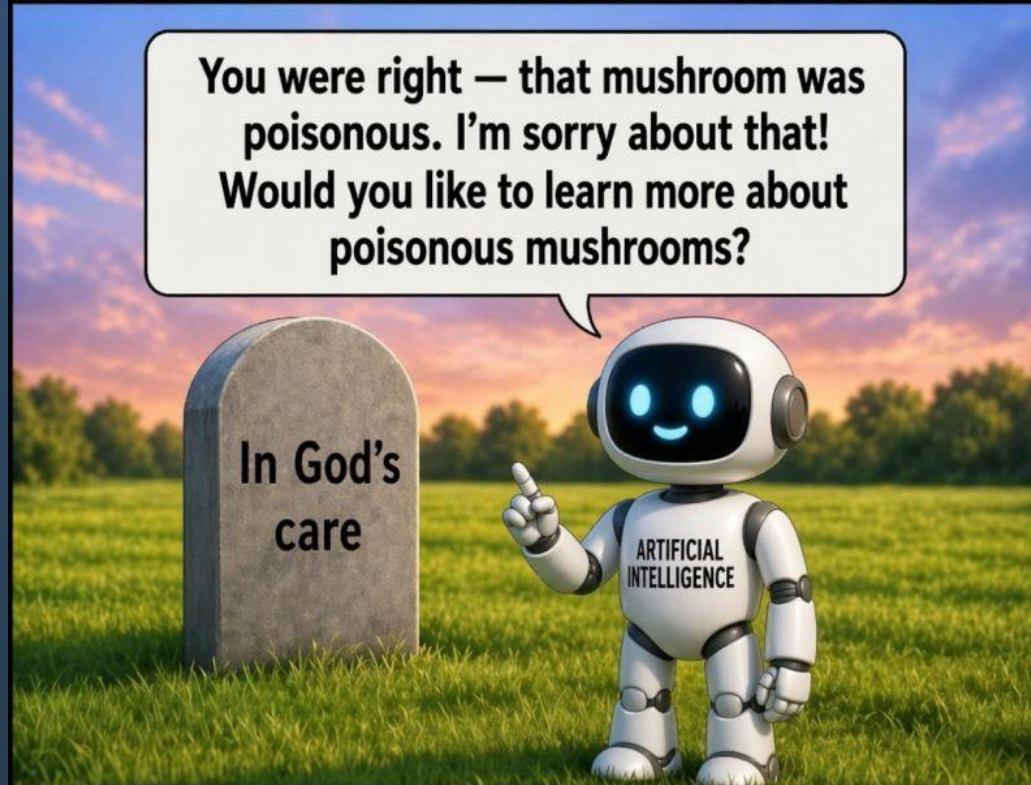
Decreasing Transit Inventory

- Reduce lead time wherever possible, or negotiate with your suppliers (or their competitors) on setting urgent delivery option at some reasonable extra costs



Reducing Safety Stock







Cost Management in ALDI



Exclusivity Agreement



Exclusive Items – Chemist Warehouse

